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Elon Musk at Tesla

The overarching purpose of Tesla Motors (and the reason I am funding the company) is to help expedite the move from a mine-and-burn hydrocarbon economy towards a solar electric economy, which I believe to be the primary, but not exclusive, sustainable solution.

— Elon Musk, 2006¹

By the end of 2022, Elon Musk had polarized the world. For some, he was the greatest entrepreneur of the early 21st century—a purpose-driven individual who not only had become enormously financially successful but also had set in motion substantial declines of carbon emissions from road transportation for decades to come. For others, Musk was the embodiment of a reckless entrepreneur who had little empathy for the risks his employees and customers were exposed to and a complete lack of appreciation for accountability and corporate governance.

Musk Early On

Musk was born in Pretoria, South Africa, in 1971. His parents divorced, and he grew up as the oldest of three siblings and two step-siblings. At first, he lived with his stepmother and father, who was active in South Africa's anti-apartheid Progressive Party.² In 1989, he moved to Ontario, Canada, with his mother and siblings, studying at Queen's University in Ontario, then transferring to the University of Pennsylvania in the United States to study economics and physics. In school, he wrote well-received papers on solar energy and supplemented his income by renting a large house and throwing parties.³

After graduating, Musk moved to Silicon Valley. He interned at energy-storage startup Pinnacle Research Institute and then at Rocket Science Games.⁴ In 1995, he began a PhD in energy physics at Stanford University but dropped out after two days. He said, "You can get a doctorate on many things that ultimately do not have a practical bearing on the world. And I really was just trying to be useful."⁵

Zippping into Business

In 1995, Musk founded Zip2 with his brother and another partner. Zip2 licensed software for newspapers to run online local city guides. Musk, who was CEO, and his co-founders raised money

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from angel investors.^{a,6} Musk reflected, “When my brother and I were starting our first company, instead of getting an apartment, we just rented a small office and we slept on the couch. [...] We showered at the YMCA and we were so hard-up that we only had one computer.”⁷ In 1996, after a failed deal to merge with top competitor CitySearch, Zip2’s board replaced Musk as CEO.⁸ In 1999, Compaq bought the company for around \$300 million, and Musk made \$22 million on the deal.⁹

His time at Zip2 demonstrated the maniacal focus, spartan lifestyle, dedication to achievement, and domineering management style that came to characterize Musk’s later endeavors.¹⁰ The taste of success fueled his hunger to create a venture that would make a significant impact on the world.¹¹ Musk soon sought a new opportunity to use the efficiencies and scale of the Internet to disrupt another industry.¹²

PayPal Payday

In 1999, Musk launched online bank X.com with three co-founders, investing \$12 million of his Zip2 earnings. He said, “Money is low bandwidth [...] it’s really just an entry in a database.”¹³ A colleague said, “That’s part of what separates Elon from mere mortals. [...] He’s willing to take an insane amount of personal risk. When you do a deal like that, it either pays off or you end up in a bus shelter somewhere.”¹⁴ A biographer wrote, “X.com held the promise of fomenting a major revolution. Musk, for the first time, would be confronting a deep-pocketed, entrenched industry head-on.”¹⁵

In 2000, X.com merged with Confinity, co-founded by venture capitalist (VC) Peter Thiel.¹⁶ Musk was CEO, but after a year of quarreling with fellow executives and employees over technical issues and much else, the board voted behind his back to replace him.¹⁷ Musk said, “It wasn’t so much that I wanted to be CEO but more like, ‘Hey, I think there are some pretty important things that need to happen, and if I’m not CEO, I’m not sure they are going to happen.’”¹⁸

In June 2001, X.com changed its name to PayPal, and Thiel became CEO. Musk still resented having the company “ripped away,” in a biographer’s words, “and given to someone else to run.”¹⁹ In 2002, eBay acquired PayPal for \$1.5 billion.²⁰ Musk was PayPal’s largest shareholder and made \$165 million from the deal, but his tenure as CEO had damaged his reputation as a leader.²¹ A biographer wrote, “[Musk] had succeeded to a large degree despite himself. Musk’s traits as a confrontational know-it-all and his abundant ego created deep, lasting fractures within his companies.”²²

Reporting and Musk’s public comments made it clear that he was aware of his shortcomings and the great potential of his talent and vision.²³ A biographer wrote, “With little prompting, Musk would start expounding on his desire to do something meaningful with his life—something lasting. His next move had to be either in solar or in space.”²⁴ In 2002, expressing a desire to do something to affect the future of human civilization, Musk left Silicon Valley and moved with his wife to Los Angeles.²⁵

SpaceX

Seeing little profit in solar technology in the early 2000s, Musk focused on his childhood passion: space exploration. A fellow PayPal executive recalled Musk spending the evening at a party closely studying “some obscure Soviet rocket manual [...] and talking openly about space travel and changing

^a Various reports claimed Musk’s father provided some initial funding for Zip2. Musk denied this, tweeting in 2019: “We started Zip2 with ~\$2k from me plus my overclocked home-built PC, ~\$5k from my bro & ~\$8k from Greg Kouri (such a good guy—he is greatly missed). My Dad provided 10% of a ~\$200k angel funding round much later, but by then risk was reduced & round would’ve happened anyway.” In the same thread, Musk denied other reports attributing early financial support to his father, writing, “He didn’t own an emerald mine & I worked my way through college, ending up ~\$100k in student debt. I couldn’t even afford a 2nd PC at Zip2, so programmed at night & website only worked during day. Where is this bs coming from?” Source: Elon Musk, Twitter.com, 5:42 and 6:22 P.M., December 28, 2019, <https://bit.ly/3Qa8i6q>, accessed August 2022.

the world.”²⁶ Efforts to study and even colonize Mars captured Musk’s attention, and he soon funded and joined the board of the nonprofit Mars Society.²⁷ In 2001, he traveled to Russia intending to buy missiles to fly mice to Mars and return them to Earth.²⁸ Around then, NASA announced it would stop missions to orbit 160 kilometers (km) to 1,000 km above the Earth’s surface.^{b,29}

When the deal in Russia fell apart, Musk decided to make his own rockets.³⁰ In 2002, he founded Space Exploration Technologies (SpaceX) with \$50 million to \$100 million of his own money.³¹ Beyond research on mice, he intended to make spaceflight cheaper by 10x and develop spacecraft for commercial flight. SpaceX’s mission was “to revolutionize space technology, with the ultimate goal of enabling people to live on other planets.”³² SpaceX touted it would offer commercial satellite-launch capabilities for \$6 million per flight, less than half the average cost at the time.³³

Burned Out: Why the World Needed Electric Cars

With SpaceX, Musk sought to ensure human civilization would survive should Earth become uninhabitable. In 2004, Musk invested in electric vehicle (EV) startup Tesla. He explained, “I think building cars is cool, but I like to be involved in things that I think are going to have a significant positive effect on the world. And I think the biggest problem terrestrially that humanity faces is sustainable energy production and consumption.”³⁴

By the time Musk invested in Tesla, scientific consensus held that human-generated greenhouse gas (GHG) emissions, particularly carbon dioxide (CO₂), were responsible for the planet’s rapid increase in average temperature. Emissions from transportation (mobility) sectors comprised between 20% and 25% of annual global CO₂ emissions.³⁵ Within mobility, the world’s 1 billion internal-combustion engine (ICE) passenger road vehicles were the leading source of emissions (around 45%) followed by road freight vehicles (29%), aviation (12%), shipping (11%), rail (1%), and other (2.2%) (see **Exhibit 1**).³⁶

Changing vehicle fuel sources was a direct avenue to slash GHG emissions. However, moving away from ICE vehicles would disrupt the incumbent auto industry, a concentrated three-tier global value chain of manufacturers, Tier 1 suppliers, and Tier 2 suppliers. Manufacturers, such as Daimler, General Motors (GM), and Toyota spent three to five years and hundreds of millions or billions of dollars to design, engineer, and market car- and truck-platforms. Manufacturers produced major components such as chassis and engines themselves, procuring the rest from Tier 1 suppliers. Tier 1 suppliers, in turn, acquired parts from Tier 2 specialists. The supply chain was tightly vertically integrated, although cross-manufacturer components supply agreements were not uncommon. (See **Exhibit 2**.)

Electric Vehicles

EVs had a long history. In 1900, only about 25% of cars used combustion power; others relied on electric batteries or steam.³⁷ Scholars wrote, “[E]arly electric vehicles provided numerous advantages over ICE vehicles: operational simplicity, no emissions of foul-smelling gases, and a quiet riding experience.”³⁸ In a battery-electric vehicle (BEV), a battery pack sent power to the transmission, which converted the power into rotational force to spin the car’s axles (see **Exhibit 3**). With fewer moving parts, BEVs produced near-instant torque without shifting through multiple gears to accelerate.

^b As it pivoted from manned, low-orbit missions, NASA announced plans to explore Mars via rovers and other un-manned means, even while struggling financially and facing some calls for the agency to close. Sources: Pat Dasch, “NASA prepares to boldly go,” *NewScientist*, October 23, 2002, <https://bit.ly/3ZpsJRU>; and Jet Propulsion Laboratory, “NASA Outlines Mars Exploration Program for Next Two Decades,” October 26, 2000, <https://go.nasa.gov/3Cs1NHj>; both accessed November 2022.

But EVs could only travel about 35 km on a single charge, and ICE vehicles, with a traveling range of hundreds of kilometers, became the norm for the next 100 years.³⁹ An important design constraint for BEVs was balancing battery capacity with total vehicle weight. Higher-capacity batteries offered diminishing returns, as the greater weight limited the distance a BEV could drive on a single charge. In addition, component materials, such as lithium, were expensive and not ecological, and larger batteries could overheat and could even explode.

Various manufacturers tried solving the issue. Amidst increased attention to fuel-efficiency due to the global energy crisis in the 1970s, Italian manufacturer Zagato sold about 500 total of its electric Zele, known in the U.S. as the Elcar.⁴⁰ The Zele could reach 30 miles per hour and had a range of 50 miles.⁴¹

Driving Down Vehicle Emissions

In 1975, the U.S. passed CAFE^c standards, and manufacturers responded by incrementally reducing per-vehicle ICE emissions. The next attempt to develop a mass-market alternative-fuel vehicle came two decades later. In 1990, California—home to nearly 30 million vehicles at the time⁴²—passed a law requiring 2% of cars in the state to be zero emissions vehicles (0EV or ZEV) by 1998, increasing to 5% by 2001, and 10% by 2003.⁴³ In 1996, GM released the EV1 to take advantage of ZEV subsidies. The BEV could drive 100 miles on a single charge of its 450-kilogram (kg) battery (around two-thirds as heavy as an average ICE car).⁴⁴ Manufacturers sued over the law, and the incentives and targets were weakened, causing GM to discontinue the EV1 after shipping fewer than 1,200 vehicles total.⁴⁵

In 1997, Toyota's Prius became the first mass-produced hybrid-electric vehicle (HEV), arriving in the U.S. market in 2000. By 2002, Toyota had delivered 100,000 Priuses cumulatively worldwide. A reporter wrote, "In the history of the automobile, there have been iconic sets of wheels, but only two cars that could truly be called groundbreaking. The Ford Model T—and the Toyota Prius."⁴⁶ In 1999, Honda launched the Insight Hybrid. In 2005, the Ford Escape Hybrid became the first U.S.-made HEV.

By the 2000s, Canada, China, the European Union (EU), Japan, South Korea, and the U.S. limited vehicle emissions on a per-km and fleetwide basis (see **Exhibit 4**). Many governments also offered manufacturer and consumer tax incentives for lower-emissions vehicles and imposed consumption taxes on fossil-fuels. Most regulations and incentives did not specify how to achieve emissions reductions, so manufacturers explored multiple approaches including hydrogen-fuel cells, BEVs, and HEV and plug-in (PHEV) vehicles with ICE powertrains to backup batteries (see **Exhibit 5**).

In 2003, Musk said, "The industry was operating under two false premises [. . .]. One, that you could not create a compelling electric car. And two, that no one would buy it."⁴⁷ Yet, the industry featured substantial barriers to entry: large capital investments in design and engineering before producing vehicles to sell; incumbents with deep pockets; tightly integrated supplier-dealer-service networks; and a physical infrastructure built around and adapted to supporting ICE transportation (e.g., gas stations).

In the Garage: Tesla's Early Days (2003–2006)

At the turn of the twenty-first century, despite increased awareness of the need to cut GHG emissions and the popularity of HEVs, BEVs were the domain of hobbyists and tinkerers. In 2003, Musk met one such engineer, J.B. Straubel. When Musk declined to invest in an electric airplane, Straubel pitched a side project: producing a marketable electric automotive powertrain using a lithium-ion

^c Corporate average fuel economy (CAFE) standards required all U.S.-produced vehicles to increase fuel-efficiency by 1985.

battery pack.⁴⁸ Musk had thought lithium was too expensive to be viable for battery packs big enough for cars, but Straubel's technology advances were promising.⁴⁹ Musk invested \$10,000.⁵⁰

Also in 2003, Martin Eberhard and Marc Tarpenning founded Tesla Motors, named for famed electricity-supply inventor Nicola Tesla. They also aimed to produce a lithium-ion battery sports car, with a prototype powered by 68,000 camcorder batteries.⁵¹ They planned to acquire the chassis and other parts from third parties, estimating production costs at \$49,000 per vehicle, 40% of which was for the battery.⁵² To break even by 2006, they estimated they needed to raise \$25 million.⁵³

The founders had seen Musk speak and thought he might be enthused by the grandeur of their endeavor. In 2004, they met, and Musk invested \$6.3 million of an initial \$6.5 million round, becoming Tesla chairman.⁵⁴ Tarpenning said, "You need angel investors to have some belief, and it wasn't a purely financial transaction for him. He wanted to change the energy equation of the country."⁵⁵

Straubel soon joined the team as a co-founder and chief technical officer. They set out to design the Tesla Roadster, powered by a battery pack with lithium-ion laptop cells and featuring a novel cooling system to prevent overheating. When completed, the high-performance car had 200 kilowatts of electrical power (288 horsepower) and accelerated from 0 to 60 miles per hour (mph) in 4.4 seconds—comparable to or slightly slower than a Porsche 911 Turbo, then one of the world's quickest ICE cars.⁵⁶

Tesla's Unique Strategic Choices

After considering whether to continue converting ICE platforms to electric powertrains, as they had with the Roadster, Tesla's co-founders decided to build future platforms from scratch. Eschewing partnerships with incumbent manufacturers and Tier 1 suppliers was a departure for a company that aimed for mass-production—a move more typical for racing cars and ultra-luxury collectors' vehicles.

At the time, although Musk spent most of his time running SpaceX, his ideas were critical to this approach. For example, in advocating to Eberhard that they build the body from high-end carbon fiber, he wrote, "Dude, you could make the body panels for at least 500 cars worth per year if you bought the soft oven we have at SpaceX! [. . .] If someone tells you this is hard, they are full of shit. [. . .] Once you've made a few of these things, you realize that there is nothing magical about glue and string."⁵⁷

Tesla also opted to distribute directly to consumers, eliminating fraught relationships with third-party dealers, who were the face of the manufacturer as far as consumers were concerned. As a Musk biographer wrote, "Lost in all of [the tension between manufacturers and dealers] was the customer, for most of whom the car-buying experience was unpleasant."⁵⁸

Struggling Off the Starting Line (2006–2009)

In 2006, Musk announced the Roadster at a fancy launch party. Tesla aimed to sell 300 per year—about 1% of the U.S. luxury sports car market—for \$80,000 per vehicle, a \$15,000 unit margin.⁵⁹ In a blog post, Musk described the Roadster as phase one of "The Secret Tesla Motors Master Plan":

So, in short, the master plan is: (1) Build sports car; (2) Use that money to build an affordable car; (3) Use *that* money to build an even more affordable car; (4) While doing above, also provide zero emission electric power generation options.⁶⁰

Musk also intended each Roadster to come with a solar-energy system from SolarCity, his cousins' solar-power system startup, of which he was chairman and lead investor.⁶¹ "This system [. . .] will

generate about 50 miles per day of electricity,” Musk wrote, “If you travel less than 350 miles per week, you will therefore be ‘energy positive’ with respect to your personal transportation.”⁶²

However, Roadster production was fraught with problems, ranging from transmission issues to trouble with the air conditioning and seats.⁶³ By 2007, Tesla had spent \$100 million, but manufacturing had not begun.⁶⁴ Costs per car were more than twice Eberhard had initially estimated.⁶⁵

As 2007 ended, Musk fired Eberhard and took over as CEO (while remaining CEO of SpaceX and chairman of SolarCity), investing \$20 million more of his own funds to keep Tesla afloat.⁶⁶ “He wasn’t ready to concede defeat, because for him there was more at stake than just creating a viable company,” a reporter wrote. “The combustion engine had made the world dependent on oil, resulting in climate change, massive transfers of wealth to oil-producing countries, and heightened geopolitical tensions. Tesla, however, could jump-start the electric car era and help solve the problem.”⁶⁷

Start Your Engines

In 2007, as Tesla fixed Roadster production issues, Musk tried to sell Tesla batteries to established automakers. In September 2007, Daimler executives agreed to visit Tesla to see the technology in action, and Musk’s team scrambled to retrofit a hastily acquired Daimler Smart Car with a Tesla battery.⁶⁸ A Daimler executive was impressed with the acceleration and torque during a test drive.⁶⁹

In 2008, Tesla released the Roadster, priced at cost at \$109,000, and sold 500 vehicles.⁷⁰ A reporter wrote, “[O]ne by one, the cars started to hit the streets. The reviews were glowing. *Car and Driver* called it a ‘revelation’ and *Motor Trend* gushed, ‘The Roadster is a cool automobile technically, a cooler automobile to drive, and an historic game-changer in our perception of battery-electric vehicles.’”⁷¹

In 2009, Tesla produced 800 Roadsters.⁷² Production costs were down to \$95,000 per vehicle, but Tesla had taken hundreds of pre-sales at a discounted price of \$92,000.⁷³ In 2009, Tesla began promoting tax incentives, explaining in a blog post, “The federal government gives tax credits of \$2,500 to \$7,500 for consumers who take delivery of an alternative-fuel vehicle after Jan. 1, 2009.”⁷⁴ Some state-level incentives waived 100% of state sales taxes, while others offered after-sale tax credits and rebates.

Yet, by the end of 2009, Tesla had less than \$500,000 in operating cash.⁷⁵ Meanwhile, Musk personally had \$20 million left from PayPal and Zip2.⁷⁶ He was going through a trying divorce, and a SpaceX rocket had recently experienced a failure during launch. Even so, Musk put the remainder of his money into Tesla and raised an additional \$40 million from his brother and other investors.⁷⁷

Revving Up

In January 2009, Daimler agreed to buy 1,000 Tesla battery packs for its Smart Cars for \$40 million.⁷⁸ Then, in May, Daimler bought nearly 10% of Tesla’s equity for \$50 million.⁷⁹ Tesla also secured \$465 million from the U.S. Department of Energy (from the same program, Ford received \$5.9 billion to develop the Focus and electrify its Transit van, and Nissan received \$1.6 billion to develop the Leaf).⁸⁰

In March 2009, Tesla announced its second vehicle, the Model S, the luxury sedan of the Secret Plan’s phase 2, which the company said would be priced at \$60,000. Within a week, customers placed more than 500 orders.⁸¹ Tesla budgeted \$130 million to design an original platform for the Model S, aiming to produce 10,000 per year.⁸² However, Tesla still assembled its vehicles in the garage of an erstwhile Los Angeles Chevrolet dealership, which doubled as a showroom.⁸³

In 2010, Musk unexpectedly succeeded in acquiring a significant auto manufacturing facility. “The car factory, known as Nummi, is located in Fremont, California, but it’s an industrial city unto itself,” a reporter wrote.⁸⁴ “It encompasses 5.5 million square feet and contains a plastics molding factory, two paint facilities, 1.5 miles of assembly lines, and a 50-megawatt power plant. Since 1984, Toyota and General Motors had run Nummi together, producing as many as 450,000 cars a year [. . .].”⁸⁵

After the great financial crisis and government bailout in 2009, GM exited Nummi. When Musk learned Toyota was shopping for a buyer at a \$1 billion valuation, he secretly toured the factory dressed as an autoworker, then promptly offered \$42 million.⁸⁶ “A month later, to his astonishment, the offer was accepted,” a reporter wrote, and Toyota invested \$50 million for a 2.5% share of Tesla’s equity.⁸⁷

Off to the Races (2010–2014)

In 2010, approximately 17,000 BEVs were on the road globally, and new sales and registrations were 0.01% of the global auto market.⁸⁸ The following year, the U.S. set a target of 1 million EVs on the road by 2015, which would save tons of annual GHG emissions per vehicle (see **Exhibit 6**).⁸⁹ In 2009, China announced a target of 500,000 HEV or EV cars and buses on the roads by 2011, which it soon updated to 2015, extending the target to 5 million EVs by 2020.⁹⁰

In 2010, Chevrolet released the Volt (HEV) and Nissan put out its LEAF (BEV). Nissan became the leading producer of EVs with 6,665 units sold that year.⁹¹ In 2010, Tesla was the smallest producer of EVs, selling a total 765 units.⁹² From 2003 to 2010, Tesla had incurred \$290 million in losses while bringing in \$150 million in revenues.⁹³

Musk decided to take Tesla public, and on June 29, 2010, Tesla’s IPO exceeded expectations, raising \$226 million, with shares closing up 40% at \$24.⁹⁴ That year, Panasonic invested \$30 million in Tesla as part of an agreement to supply lithium-ion batteries for Tesla’s vehicles.⁹⁵ In 2011, the companies extended the deal and announced investments into battery technology advancements.⁹⁶

Model S

By 2012, Tesla had sold 2,500 Roadsters.⁹⁷ As the Model S launch approached, Musk established a hardworking workplace culture. In an email titled “Ultra Hardcore,” he told employees:

Please prepare yourself for a level of intensity that is greater than anything most of you have ever experienced before [. . .]. Revolutionizing industries is not for the faint of heart, but nothing is more rewarding or exciting, and I will personally make sure that those who produce exceptional results are rewarded exceptionally, as is fair and right.⁹⁸

He followed up with a second email telling employees to warn their families and friends “they will see much less of you for the next six months and there will be almost zero vacation. [. . .] Please reduce time off and travel plans to the absolute minimum necessary to maintain sanity and avoid divorce.”⁹⁹

However, later that year, the Model S earned a tepid reception at launch. One observer wrote:

Even considering the supposed pent-up demand among environmentalists for a car like the Model S, [Tesla has] ambitious goals for a small company planning to launch a niche luxury product into a soft market. Frankly, we’re skeptical. We’ve seen how brutal and unforgiving the market can be, and other automakers aren’t simply going to roll over and surrender that volume to Tesla. Nissan is leading the way with its bargain-priced Leaf

[sic], and the Chevy Volt, although a different breed of vehicle, will compete almost directly with the Model S at half the price.¹⁰⁰

Musk's team designed the Model S with no exhaust pipes, catalytic converter, nor thermal shielding, since there was little engine heat (70% of the energy produced by combusting gasoline in an ICE was wasted as engine heat, while 10% to 20% powered the car).¹⁰¹ Without those artifacts of ICE design, the vehicle had more cabin and storage space than most luxury sedans, including enough room for a third row of seats, to seat seven people. The Model S was also faster than its peers, accelerating from 0 to 60 mph in 4.2 seconds, with a top range of 300 miles on a single charge, although the basic model was certified to only 160 miles. An upscale Signature Model S, outfitted with a larger battery, was certified to 225 miles but could cost well over \$100,000, depending on features.¹⁰²

The Model S also included novel amenities, such as a 17-inch touchscreen center console, with GPS navigation and Internet-connected information and entertainment features. A flat battery pack, containing 7,000 lithium-ion batteries and weighing 1,300 pounds, was located under the floorboard.¹⁰³ It took 10 hours to fully charge, with free charging at Tesla Supercharging stations, which were being installed all over the U.S. (see **Exhibit 7**).¹⁰⁴ When vehicles needed service, Tesla could often fix the issue through an Internet-connected software repair, rather than having to take the car to a mechanic.¹⁰⁵

Said an early Model S owner, "It changes everything about transportation. It's a computer on wheels."¹⁰⁶ A reporter wrote, "Musk didn't just want to build a functioning electric sedan—he wanted to redefine what a sedan could be."¹⁰⁷

Retailing the Model S

Once again, Tesla pivoted from industry convention with its marketing strategy. Sticking to its no-dealership approach, Musk took inspiration from the Apple Store. In 2012, Tesla opened seven showrooms in high-end shopping malls with high foot traffic (see **Exhibit 8**).¹⁰⁸ In stores, Tesla displayed the Model S and its two-square-foot electric motor, along with touch screens where customers could see analysis of EV vs. ICE emissions and ownership costs.

Store employees received no commissions.¹⁰⁹ "Our goal instead is to interact with people when they're looking at new things in the mall and not necessarily in the market for a car. The focus is on answering the many questions they have about electric vehicles and the Tesla Model S," said Tesla's head of retail.¹¹⁰ When customers decided to buy, they could place orders at stores or online. Tesla offered limited customization, including few color choices, and customers configured options themselves via a self-service sales system. Customers could either pick up vehicles from stores or have them delivered at home—an approach other manufacturers and dealers quickly mimicked.

In 2012, Tesla sold around 3,000 Model S units, short of its target of 5,000.¹¹¹ Production and order-management issues led to delivery delays, and some features did not work on some early deliveries.¹¹² Some customers complained about fit-and-finish issues compared to the standards among luxury-priced peers like Audi or BMW.¹¹³ Tesla targeted delivering 20,000 Model S units by the end of 2013.¹¹⁴ Yet challenges continued, as the press covered four occasions from October 2013 to February 2014 when Model S batteries caught fire after being damaged in separate accidents.¹¹⁵

Still, the car became a sensation. In May 2013, *Consumer Reports* named the Model S the best car it had ever tested, giving it a score of 99 out of 100.¹¹⁶ That year, the Model S also won the prestigious *Motor Trend* Car of the Year title.¹¹⁷ Tesla delivered 22,477 Model S vehicles in 2013.¹¹⁸

Also in 2013, Tesla renewed its agreement for Panasonic to supply two billion battery cells by 2017.¹¹⁹ Shortly after, Musk announced Tesla would also begin producing its own lithium-ion batteries in the soon-to-be-built Gigafactory 1 in Storey County, Nevada.¹²⁰ Musk said, “This is going to be a very green factory. There’s going to be a lot of solar power. It’s going to have essentially zero emissions and there are no toxic elements that are going to come out.”¹²¹ Recycling battery packs for use in new vehicles would be built into the factory’s design.¹²² The following year, Panasonic announced it would invest 30% to 40% of the 10 million square-foot factory’s \$5 billion total costs.¹²³

Fires, Frustrations, and Fatalities (2015–2018)

By 2015, the Model S was still the only electric car on the market that could drive more than 100 miles on a single charge. Its leading competitor was the Nissan LEAF, which sold 17,269 units in 2015.¹²⁴ Yet, Tesla was losing \$4,000 on each vehicle sold, and, despite growing sales, a series of problems challenged the company’s reputation over the next few years.¹²⁵

An X against the Model X

In September 2015, Tesla began delivering its long-delayed next vehicle, the Model X luxury SUV.¹²⁶ Customers paid \$132,000 or more for a Model X, which featured double-hinged falcon wing doors that opened vertically instead of horizontally.¹²⁷ In October 2015, Tesla released a software update that introduced self-driving automation features (the driver’s attention and control were still required at all times).¹²⁸ In 2016, Musk claimed that by 2018 Tesla’s autopilot feature would be able to self-navigate a vehicle from New York to Los Angeles – and more efficiently and safely than a human driver.¹²⁹

In early 2016, Model X malfunctions started occurring, including doors that failed to open.¹³⁰ Tesla voluntarily recalled 2,700 Model X units for an issue with the third-row seat latch.¹³¹ In 2016, Tesla filed a lawsuit against the hydraulics supplier for the falcon-wing doors, which “leaked oil and sagged or produced excessive heat.”¹³² The case was settled for an undisclosed sum in September 2016.¹³³

In 2016, Tesla released its \$35,000 Model 3, representing the final phase of the “The Secret Tesla Motors Master Plan.”¹³⁴ Hundreds of people waited at Tesla locations all over the world to place orders.¹³⁵ In its first week, there were more than 232,000 pre-orders.¹³⁶ That year, Tesla also announced its European expansion with the purchase of Germany’s Grohmann Engineering.¹³⁷

Autopilot Accidents

In May 2016, a Model S driver died in a fatal crash while using Tesla’s autopilot software.¹³⁸ Tesla soon updated the software to alert the driver to return their hands to the wheel after a shorter interval.¹³⁹ The U.S. National Highway Transportation Safety Administration (NHTSA) found that Tesla was not responsible for the crash but recommended modifying the airbag settings for autopilot.¹⁴⁰

Later that year, Tesla began charging vehicle owners \$8,000 to access its full self-driving and autopilot capabilities.¹⁴¹ In November, Tesla released a promotional video purportedly showing its autopilot feature in action for over two minutes of autonomous driving. A class-action lawsuit alleged the video was fraudulently edited and resulted in a \$5.4 million settlement with Tesla.¹⁴²

In March 2018, a fatal auto accident involving a Model X occurred in Mountain View, California. The crash apparently resulted from glitches with Tesla’s autopilot system that the driver had previously reported.¹⁴³ According to the press, an NHTSA report determined, “Autopilot failed to keep

the driver's vehicle in the lane, that its collision-avoidance software failed to detect a highway barrier and that the driver was probably distracted by a game on his phone."¹⁴⁴

Shareholders Approve the Controversial SolarCity Acquisition

The affordable Model 3 squeezed Tesla's margins. As 2016 closed, the company had \$1.4 billion of free cash on hand. If it continued spending \$500 million per quarter, on average, it would run out of cash before the end of the following year.¹⁴⁵ The company had \$3.4 billion in debt, and its stock price had dropped by more than three-quarters since its highest point in February 2014.¹⁴⁶

Yet, in November 2016, after months of discussion, Tesla bought SolarCity for \$2.6 billion. Musk recused himself from the vote, and 85% of Tesla shareholders approved.¹⁴⁷ The previous month, when rallying support for the deal and showcasing a new home-solar roof tile and Tesla's home-powering Powerwall battery, Musk said, "This is sort of the integrated future. An electric car, a Powerwall and a solar roof. The key is it needs to be beautiful, affordable and seamlessly integrated."¹⁴⁸

Yet, many investors and observers criticized the deal. Musk was the largest shareholder of both companies, other directors sat on both boards, and his cousins still ran SolarCity, which had lost half its market capitalization over the previous year.¹⁴⁹ An investor who shorted both companies called the acquisition a "shameful example of corporate governance at its worst."¹⁵⁰ Analysts faulted Tesla for overpaying for a company outside its core operations, exposing the automaker to new competitive and regulatory risks. Tesla's stock dropped more than 10% when the deal was announced.¹⁵¹

Shareholders Approve Musk's Controversial Compensation Plan

In January 2018, Tesla's board endorsed a proposal to award Musk a stock grant that could earn him up to \$55.8 billion in stock and awards if Tesla's market cap reached \$650 billion.¹⁵² The vote passed by a wide margin. At the time, Tesla's market cap was around \$52 billion. Musk and his brother Kimbal recused themselves from the vote.¹⁵³

Musk would receive no other compensation for his work at Tesla outside this plan. The grant gave Musk \$2.6 billion in stock options in 12 tranches that each vested over 10 years. Tesla had to reach a market cap of \$100 billion for the first tranche to vest, and then each of the remaining 11 tranches required an additional \$50 billion in market value. In addition, there were 16 operational milestones: eight focused on increasing revenue from \$20 billion up to \$175 billion, and another eight targeting adjusted EBITDA milestones ranging from \$1.5 billion to \$14 billion.¹⁵⁴

Several analysts and corporate governance experts reacted negatively to the compensation plan. Powerful proxy advisors ISS and Glass Lewis both recommended rejecting the proposal because the incentives were excessive and Musk, who already owned 20% of Tesla's stock, was already incentivized to perform his duties. Efraim Levy, an analyst with research investment firm CFRA, wrote, "We question why Musk has to be incentivized so highly given his high equity interest in TSLA and risks to his personal reputation. We view TSLA shares as expensive at current valuations, despite our forecast of 2019 profitability."¹⁵⁵ Others supported the compensation plan, suggesting it aligned incentives between shareholders and Musk and would keep Musk focused on Tesla rather than his other endeavors.¹⁵⁶ Yet others provided a different viewpoint:

The design of the compensation plan and its announcement were not about compensation at all. They were about signaling a credible commitment to Tesla's purpose: to become a clean energy giant. [. . .] To get there, Musk needs not only the normal sort of investor confidence, but for investors to buy into his radical vision for the company. You

don't invest in Tesla for safe returns, but for the possibility of it becoming one of the century's most important companies.¹⁵⁷

Investors Short the Tesla Stock

By 2018, some investors began to doubt Tesla could scale Model 3 production to meet demand without further infusions of capital.¹⁵⁸ Investors also worried about vehicle safety and the viability of the Tesla brand after the Model X fatal crash. By April 2018, around 25% of available Tesla shares were sold short, up 10% over the previous month, and some analysts recommended selling.¹⁵⁹ Prominent research analysts, including one from Goldman Sachs, recommended selling.¹⁶⁰

On August 7, 2018, Musk tweeted, "Am considering taking Tesla private at \$420. Funding secured," followed by, "Shareholders could either to sell at 420 or hold shares & go private."¹⁶¹ Tesla's share price spiked 11%, costing short sellers \$1.3 billion.¹⁶² The following week, Musk criticized short-sellers in a *New York Times* interview, complaining they contributed to a personally stressful year.¹⁶³ Tesla shares fell 9% the next day, generating \$1 billion in value for the short-sellers.¹⁶⁴

SEC Investigation

On September 27, 2018, the U.S. Securities and Exchange Commission (SEC) charged Musk with securities fraud, referring in a press release to his "series of false and misleading tweets about a potential transaction to take Tesla private."¹⁶⁵ The statement noted:

On August 7, 2018, Musk tweeted to his 22 million Twitter followers that he could take Tesla private at \$420 per share (a substantial premium to its trading price at the time), that funding for the transaction had been secured, and that the only remaining uncertainty was a shareholder vote. The SEC's complaint alleges that, in truth, Musk had not discussed specific deal terms with any potential financing partners, and he allegedly knew that the potential transaction was uncertain and subject to numerous contingencies. According to the SEC's complaint, Musk's tweets caused Tesla's stock price to jump by over six percent on August 7, and led to significant market disruption.¹⁶⁶

Tesla was not spared. "[D]espite notifying the market in 2013 that it intended to use Musk's Twitter account as a means of announcing material information about Tesla and encouraging investors to review Musk's tweets, [Tesla had] no disclosure controls or procedures in place to determine whether Musk's tweets contained information required to be disclosed in Tesla's SEC filings," the SEC wrote. "Nor did it have sufficient processes in place to [ensure] Musk's tweets were accurate or complete."¹⁶⁷

On September 29, 2018, the SEC announced that it had reached a settlement. Tesla and Musk agreed to pay \$20 million each in penalties, to be given to harmed investors.¹⁶⁸ Musk also agreed to resign as Tesla chairman for at least three years, to be replaced by an independent chairman. In addition, Tesla had to appoint two new independent board directors and create a committee and additional controls and procedures to oversee Musk's communications.¹⁶⁹

On October 4, 2018, Musk joked about the SEC's acronym, tweeting, "Just want to [say] that the Shortseller Enrichment Commission is doing incredible work. And the name change is so on point!"¹⁷⁰ In November 2018, Robyn Denholm, CFO of Juniper Networks, a four-year Tesla board veteran who had chaired the audit committee, became chair of Tesla's board. Larry Ellison, co-founder of Oracle, and Kathleen Wilson-Thompson, executive vice president of Walgreens Boots Alliance, joined the board as its two new independent directors (Ellison left the board in August 2022).¹⁷¹

Sales Sputtering

Tesla closed 2018 with nearly 250,000 deliveries, more than double 2017's total but less than half of the half-million car target set by Musk in 2016.¹⁷² To hit a target of 7,000 Model 3s produced per week, Musk kept some workers over the Thanksgiving holiday (on a volunteer basis).¹⁷³ Tesla opened 2019 by limiting new production to filling reservations in China and Europe.¹⁷⁴ In January, the company fired 7% of its workers.¹⁷⁵ In March, when Musk announced Tesla would shutter all but a few of its 100+ retail locations – Tesla having just opened 27 locations in the final quarter of 2018 – the share price fell 13%.¹⁷⁶ Musk claimed the overwhelming majority of Model 3 orders were placed online.¹⁷⁷

Meanwhile, short sellers kept betting Tesla's stock would crash. Short seller Andrew Left, founder of Citron Research, noted in a lawsuit against Tesla, "Musk has a long-standing public feud with short-sellers and often uses his personal Twitter account to taunt and confront skeptics of his company."¹⁷⁸ He went on to suggest Musk should relinquish the CEO position.¹⁷⁹ Long-standing Tesla short seller and hedge fund manager David Einhorn, founder of Greenlight Capital, who rose to fame for a prescient call on Lehman Brothers' accounting trouble before its collapse, compared Tesla to Lehman Brothers, predicting that it would also soon fail.¹⁸⁰ He later wrote, "The signs are everywhere, from the lack of demand, desperate price cutting, layoffs [. . .] rushed product announcements and a new effort to distract investors from the demand problem with hyperbole over TSLA's autonomous driving capabilities. TSLA has lost a significant number of senior executives and appears to be having a hard time recruiting replacements. After all, who would want to work in such an environment?"¹⁸¹

As Tesla's market cap rose, leading short sellers to take heavy losses, Musk mocked them, calling short selling a scam. He sent Einhorn a box of shorts and even went on to advertise red satin shorts, tweeting, "Limited edition short shorts now available," via the Tesla website.¹⁸²

Fully Charged: The EV Market Grows

Despite its setbacks and challenges, many credited Tesla with jumpstarting the global EV market. A reporter wrote, "Tesla is forcing the auto industry to change rapidly. Tesla didn't invent the electric car (Scottish inventor Robert Anderson did, in 1832), but it was Tesla who popularized, pioneered and promoted the electric car ever since the company's founding in 2003."¹⁸³ Tesla aimed to increase sales to 500,000 units annually as of 2018.¹⁸⁴ Yet other manufacturers were catching up. BMW's 2017 target was to sell 100,000 EVs, representing about 15% to 25% of its products.¹⁸⁵ As of 2016, Volkswagen planned on selling 3 million EVs per year by 2025, representing one-quarter of its total sales.¹⁸⁶

Following the 2015 Paris Climate Accord, many governments moved to accelerate EV adoption (see **Exhibit 9**). In 2017, France announced it would prohibit sale of ICE vehicles starting in 2040.¹⁸⁷ The U.K. soon followed suit.¹⁸⁸ In September 2017, China announced it would ban ICE vehicles, as well, with restrictions set to take effect at different times in different regions.¹⁸⁹

Technology improvements also meant EV ownership costs were falling. Lithium-ion battery prices dropped by half from 2008 to 2013 – and by 97% since the 1980s – and analysts expected continued declines.¹⁹⁰ Cumulative demand soared from zero gigawatt hours (GWh) in 2010 to around 500 GWh by 2020, with projections to increase exponentially to 2,000 GWh by 2024 and 9,300 GWh by 2030.¹⁹¹ (See **Exhibit 10**.) Experts estimated that BEVs needed to reach \$100 to \$150 per kilowatt hour (kWh) to achieve price parity with ICE vehicles.¹⁹² In 2016, Tesla, the industry leader, estimated its batteries cost at \$190 per kilowatt hour to produce, down from \$1,000 in 2010; the industry average was \$227.¹⁹³

Shifting into High Gear (2019–2021)

Despite losing \$702 million in Q1 2019, Tesla closed the year with \$24 billion in revenue (see **Exhibit 11**).¹⁹⁴ The stock price rebounded from a three-year low of \$180 to finish 2019 at a then-all-time high over \$400.¹⁹⁵ The company also unveiled the upcoming Model Y compact SUV, which would be priced between the Model 3 and Model X, and Cybertruck, Tesla's take on a pickup truck.¹⁹⁶ In November, Tesla announced a new Gigafactory to be built in Germany.¹⁹⁷ In 2020, Tesla delivered 500,000 vehicles.¹⁹⁸

In 2020, global EV sales beat forecasts, increasing by two-thirds over 2019, while overall auto sales fell 16% due to the COVID-19 pandemic.¹⁹⁹ That year, the total cost of ownership for an EV was lower than that of an ICE vehicle in Europe and the U.S. (worldwide upfront price parity was expected in 2022 to 2024).²⁰⁰ Worldwide, in 2020, there were 1.3 million installed public EV charging stations, set to increase to 1.8 million the following year.²⁰¹ Over 1.1 million of the charging stations were in China, the U.S. was committed to having 500,000 by 2025, and the EU aimed for 1 million by the same time.²⁰²

Richest Person in the World

In January 2021, Tesla's market value hit \$700 billion. Musk's net worth was \$185 billion, making him the richest person in the world.²⁰³ An analyst wrote, "The owner of a business which has only just made its first annual profit and is a minnow compared to the likes of Toyota—or Amazon—is now the richest person."²⁰⁴ Musk tweeted: "About half my money is intended to help problems on Earth, and half to help establish a self-sustaining city on Mars to ensure continuation of life (of all species) in case Earth gets hit by a meteor like the dinosaurs or WW3 happens and we destroy ourselves."²⁰⁵

In Q2 2021, Tesla delivered a record number of vehicles, despite microchip and other parts shortages plaguing the entire industry.²⁰⁶ Yet Musk urged Tesla employees to push harder. In September, he emailed employees, telling them to "go super hardcore" to "ensure a decent Q3 delivery number" amidst "the biggest wave in Tesla history," saying, "we got to get it done."²⁰⁷ In October of 2021, rental company Hertz agreed to purchase 100,000 Tesla Model 3s for \$4 billion in 2022.²⁰⁸ On the news, Tesla's stock rose 12%, topping \$1 trillion in market cap.²⁰⁹ On December 21, 2020, S&P added Tesla to the S&P 500, and, in May 2021, to its S&P 500 ESG Index, the broadest S&P index that scored equities on their environmental, social, and governance (ESG) profiles.^{d,210}

Birdwatching

In January 2022, Musk began inquiring privately about buying Twitter, the influential 70 million-user microblogging social media site with \$5 billion in 2021 revenues. This, again, raised governance concerns about Musk's becoming distracted from his responsibilities at Tesla. By March, he was the largest Twitter shareholder, with 9% of its equity, worth around \$2.9 billion.²¹¹ In early April, Twitter announced Musk would join its board; a few days later, he announced he would not.

On April 14, 2022, Musk filed with the SEC his intention to acquire Twitter for \$54.20 per share, or approximately \$44 billion, a 38% premium over the day's share price.²¹² Musk, a self-declared political

^d The S&P 500 ESG Index was "a broad-based, market-cap-weighted index that is designed to measure the performance of securities meeting sustainability criteria, while maintaining similar overall industry group weights as the S&P 500." Historically, the performance of S&P 500 ESG Index had closely tracked the S&P 500. Sources: Margaret Dorn, "The (Re)Balancing Act of the S&P 500 ESG Index," Indexology Blog, S&P Dow Jones Indices, a Division of S&P Global, May 17, 2022, <https://bit.ly/3G9IyD0>, accessed May 2022; S&P 500 Index, S&P Dow Jones Indices, <https://bit.ly/3iSJIAK>, accessed December 2022.

centrist wrote, “I invested in Twitter as I believe in its potential to be the platform for free speech around the globe, and I believe free speech is a societal imperative for a functioning democracy.”²¹³

On October 28, 2022, after much drama, during which Musk had attempted to withdraw from the deal, on October 26, he visited Twitter’s offices carrying a porcelain sink, tweeting, “Entering Twitter HQ—let that sink in!” On October 28, he formally acquired Twitter. On taking over, Musk fired CEO Parag Agrawal, installing himself as the top executive with the title “chief twit.” In early November 2022, via email, Twitter informed thousands of workers they were fired, eventually culling around 50% of Twitter’s 7,500 employees (some fired employees were evidently hired back).²¹⁴

On November 15, 2022, at midnight, Musk sent all employees an email: “Going forward, to build a breakthrough Twitter 2.0 and succeed in an increasingly competitive world, we will need to be extremely hardcore. This will mean working long hours at high intensity. Only exceptional performance will constitute a passing grade.”²¹⁵ Musk gave employees until 5:00 p.m. the following day to either commit to the new culture or accept three months’ severance.²¹⁶ At Tesla, Musk had likewise recently announced a strict return-to-office policy following a period of hybrid and work-from-home arrangements during COVID-19.²¹⁷ Less than half of employees had committed to Twitter 2.0 by the deadline, and hundreds of others quit.²¹⁸

EV and ESG

In 2022, analysts projected that, by 2030, automakers worldwide would spend more than \$500 billion manufacturing EVs.²¹⁹ There were dozens of EV models on the market, ranging in price from the mid-\$20,000s and up. Across models, battery performance and price had improved significantly over the previous decade, reducing emissions dramatically compared to ICE vehicles (see **Exhibit 12**).

However, the environmental toll of extracting critical minerals—especially lithium—and emissions from manufacturing counteracted some of an EV’s emissions savings. A climate scientist explained, “Electric and conventional vehicles have similar manufacturing emissions with the exception of the battery. Excluding the battery, it takes ~5.7 tons of CO₂ emissions to manufacture the vehicle [. . .]. Battery manufacturing is quite emissions intensive [. . .]. [R]esearch suggests a central estimate of about 84 kg CO₂ per kWh battery capacity.”²²⁰ Even so, EVs could reach emissions parity with ICE vehicles after about 15,000 miles of use, and over their lifecycles could emit 3x less GHGs than ICE.²²¹

ESG ratings frameworks to assess auto manufacturers attempted to account for such complexities. Environmental rubrics included factors such as product carbon footprint, toxic emissions and waste, raw materials sourcing, and carbon emissions.²²² Social and governance factors included product safety and quality, labor management, business ethics, pay, and tax transparency. (See **Exhibit 13**.)

Tesla and ESG

ESG rating methodologies played to Tesla’s detriment. In May 2022, S&P removed Tesla from its S&P 500 ESG index. S&P Dow Jones Indices’ head of ESG indices for North America Margaret Dorn cited, “Tesla’s (lack of) low carbon strategy and codes of business conduct, claims of racial discrimination and poor working conditions at Tesla’s Fremont factory, as well as its handling of the NHTSA investigation after multiple deaths and injuries were linked to its autopilot vehicles. While Tesla may be playing its part in taking fuel-powered cars off the road, it has fallen behind its peers when examined through a wider ESG lens.”²²³ Another influential financial institution, MSCI, ranked Tesla as “average among 43 companies in the automobiles industry” in its ESG rating.²²⁴

Musk was outraged by such analyses. He tweeted, “Exxon is rated top ten best in world for environment, social & governance (ESG) by S&P 500, while Tesla didn’t make the list! ESG is a scam. It has been weaponized by phony social justice warriors.”²²⁵ Tesla’s 2021 Impact Report echoed Musk, stating, “Current environmental, social and governance (ESG) reporting does not measure the scope of positive impact on the world. Instead, it focuses on measuring the dollar value of risk/return,” continuing, “We need to create a system that measures and scrutinizes actual positive impact on our planet.”²²⁶

The Road Ahead

In 2022, the future looked bright for EVs. The global EV fleet was estimated at 20 million vehicles, demonstrating strong growth since 2015.²²⁷ China and Europe accounted for 46% and 34% of EV sales, respectively.²²⁸ The U.S. secured third place, with 15% of total sales.²²⁹ In 2021–2022, China, the EU, and the U.S.—where more than 90% of global EV sales occurred—each put in place ambitious new targets for phasing out ICE vehicles.²³⁰

Tesla was by far the most valuable auto manufacturer in the world by market cap (see **Exhibit 14**). Leaders also included China’s SAIC, Daimler, Hyundai, and Volkswagen.²³¹ In 2022, analysts estimated 71 million EVs would be on the road globally by 2025, 10 years ahead of 2016 estimates.²³²

On September 30, 2022, Musk revealed a prototype of Tesla’s humanoid robot ‘Optimus.’ He predicted Optimus would outgrow Tesla’s EV business and the company would sell millions over the next decade, for under \$20,000 each, and would begin taking orders in three to five years.²³³ Musk said, “I think the mission does somewhat broaden with the advent of Optimus to [. . .] making the future awesome.”²³⁴ On December 1, Tesla made the first deliveries of its years-delayed heavy-duty truck to PepsiCo, five years after it was first unveiled, boasting an unusual design with the driver sitting in the center of the cab, a faster acceleration than diesel-powered trucks, and 500 miles of range on a single charge.²³⁵

However, with the Federal Reserve raising interest rates drastically in 2022, the stock valuations of high-growth companies, such as Tesla, came down sharply, and Tesla experienced a decline of more than 60%.²³⁶ Meanwhile, Twitter was losing \$4 million a day according to Musk.²³⁷ These losses allowed him to collect another accolade: the first person to lose more than \$200 billion in wealth.²³⁸

Against this backdrop, some called for Musk to step down as CEO of Tesla. The third-largest individual-shareholder of Tesla tweeted, “Elon abandoned Tesla and Tesla has no working CEO. Tesla needs and deserves to have working full time CEO. What Tesla BOD (board of directors) should do, do nothing?”²³⁹ Musk seemed to agree that he should be accountable to Tesla shareholders, saying that it was important that they had a role in vetting the company’s wide-ranging efforts: “If I go crazy, you can fire me. This is important.”²⁴⁰

In January of 2024, after the Tesla stock appreciated during 2023 by more than 100% to recover much but not all the 2022 losses, Musk tweeted, “I am uncomfortable growing Tesla to be a leader in AI & robotics without having ~25% voting control. Enough to be influential, but not so much that I can’t be overturned.”²⁴¹ He currently held about 13% of the shares and the votes in Tesla. Should investors support his request for increased influence? Should Tesla board members grant his wish?

Exhibit 1 CO₂ Emissions by Mode, gigatons (Gt), 2000 to 2030 (estimated)

	Passenger Road Vehicles	Road Freight Vehicles	Aviation	Shipping	Rail	Other
2000	2.54	1.70	0.67	0.62	0.09	0.14
2005	2.88	1.90	0.73	0.70	0.10	0.16
2010	3.12	2.08	0.74	0.80	0.09	0.16
2015	3.48	2.29	0.87	0.82	0.09	0.16
2020	3.64	2.41	0.94	0.86	0.08	0.17
2025	3.52	2.43	0.90	0.85	0.07	0.17
2030	3.11	2.30	0.87	0.83	0.06	0.16

Source: International Energy Agency, "Transport sector CO₂ emissions by mode in the Sustainable Development Scenario, 2000–2030," IEA, Paris, last updated January 5, 2022, <https://bit.ly/3WMyIPn>, accessed May 2022.

Note: "Other" includes pipeline and non-specified transport.

Exhibit 2 The Auto Industry

In the 20th century, the "Big 3" U.S. automakers, Chrysler, Ford, and General Motors (GM), along with Europeans, such as Daimler and Volkswagen, were the largest manufacturers. Since the 1980s, the industry had globalized. In 2003, the year Tesla was founded, manufacturers produced and sold more than 60 million cars worldwide; in 2019, that number reached nearly 80 million, before falling to 68 million by 2022. Manufacturers marketed variations of platforms in mass-market, premium, and luxury segments. For example, Toyota marketed the same underlying components as the Corolla, Camry, and the premium Lexus ES. Higher segments featured more powerful drivetrains, better safety features, and more amenities, such as fancy "infotainment" systems.

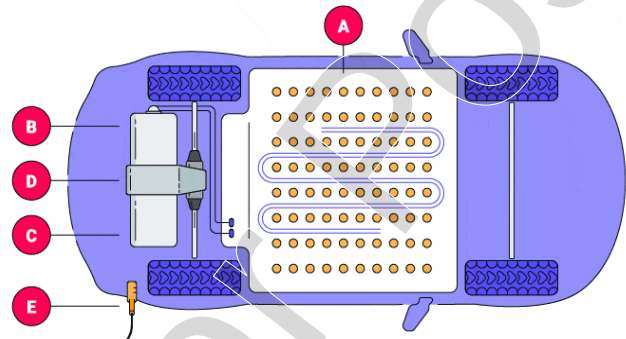
In total, the auto industry spent about 5% of revenues on advertising and marketing. Most manufacturers relied on third-party franchised dealer networks to bring vehicles to market and sell. In the early and mid-20th century, the Big 3 manufacturers benefited from exclusive arrangements that made dealers dependent on them. Pushing back over several decades, dealers became a powerful lobbying bloc in their own right, leading to regulations across the U.S., limiting how and where manufacturers and dealers could sell vehicles. In the 21st century, the dealership model remained the prevailing norm for distribution in the U.S. and internationally.

	Dec. 2003 Market Cap	2003 Total Vehicles Produced
GM	\$75 billion	8.2 million
Toyota	\$94 billion	6.8 million
Ford	\$57 billion	6.7 million
Daimler/Chrysler	\$47 billion	4.3 million
Volkswagen	\$15 billion	4.3 million
Peugeot/Citroen	\$12 billion	3.3 million

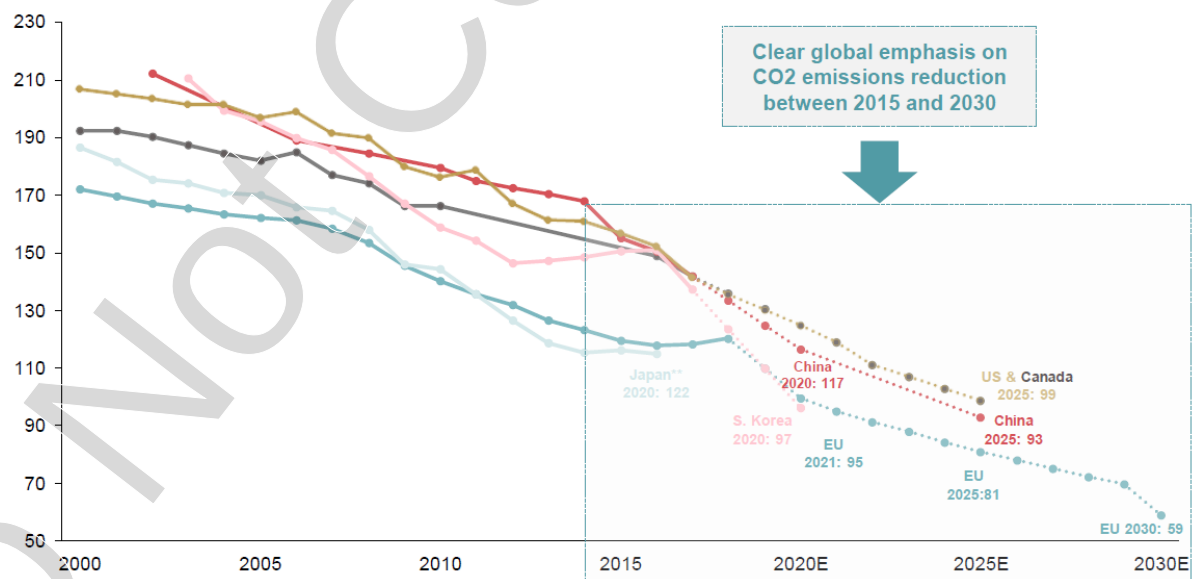
Source: Compiled by casewriter from "Global vehicle production and sales by manufacturer," *Automotive News*, 2005 Market Data Book, <https://bit.ly/3G8vU94>; Neil Winston, "Global Car Sales To Fall, Spooked By Russian Invasion, China's Shutdown," *Forbes*, May 17, 2022, <https://bit.ly/3fWyEvE>; Nathan Gomes, "Auto industry ad spend falls to \$4.8 billion in first seven months of 2022, data show," Reuters, September 16, 2022, <https://reut.rs/3fW6mRZ>; and IBISWorld, "Car & Automobile Manufacturing in the US—Market Size 2005–2028," September 24, 2022, <https://bit.ly/3WVb6bd>, all accessed November 2022; and company entries in Stockanalysis.com and Companiesmarketcap.com, accessed January 2023.

Exhibit 3 Primary Components of an Electric Car

- A: Battery packs produce direct current power and are made of collections of about 7,000 common lithium-ion cells (batteries) with coolant running through the battery packs to prevent overheating.
- B: Inverters convert direct current power to alternating current, which powers the motor.
- C: Alternating current power produces a rotating magnetic field in the induction motor, causing it to turn.
- D: Drivetrains in EVs have single-speed transmissions that distribute power to the wheels.
- E: Charging EVs works by plugging them in at home (120 volts (v) or 240v; 2–5 miles per hour of charge time); at public charging stations (120v/240v; 10–25 miles per hour of charge time); or at direct current fast-charging stations (~80% of full charge in 30 minutes of charge time). Using regenerative braking, EVs also converted kinetic energy from deceleration into electricity through the motor, storing it in the battery pack.



Source: Susan Meyer, "How an Electric Vehicle Works," Zebra, July 7, 2022, <https://bit.ly/3loa3vG>, accessed January 2023.

Exhibit 4 New Passenger Car Emissions Limits, grams of CO₂ per km, 2000 to 2030e, as of 2019

Source: Stephen Reitman and Steve Pereira Fernandes, "Automotive Sector: Coping with Disruption and Economic Uncertainties," Société Générale, September 2022, p. 26, via Refinitiv/AMR, accessed November 2022.

Note: Normalized to the New European Driving Cycle (NEDC), an old driving cycle designed to assess the emission levels of car engines and fuel economy in passenger cars, first conceived in the 1980s and fatally undermined by manufacturer manipulation or "optimization." It was replaced on September 1, 2018, by the more rigorous Worldwide Harmonized Light Vehicles Test Procedure (WLTP), which gives higher CO₂ readings than the NEDC system.

Exhibit 5 Types of Electric Vehicles

Battery Electric Vehicle (BEV) A BEV was an all-electric vehicle powered by electricity stored in a battery pack and delivered to the electric engine. BEVs solely ran on electricity and did not have gas engines. They had to be plugged into an external power source to recharge and also recharged via regenerative braking.

Plug-In Electric Vehicle (PHEV) A PHEV ran on an electric motor powered by a battery pack but also had an ICE that required fuel. They can be plugged into a power source or use regenerative braking to recharge.

Hybrid Electric Vehicle (HEV) An HEV had a battery pack that recharged through regenerative braking but could not be plugged in to charge. HEVs start with electric motors, and the ICE turned on and off as needed. Both motors were controlled by an internal computer that monitored driving conditions.

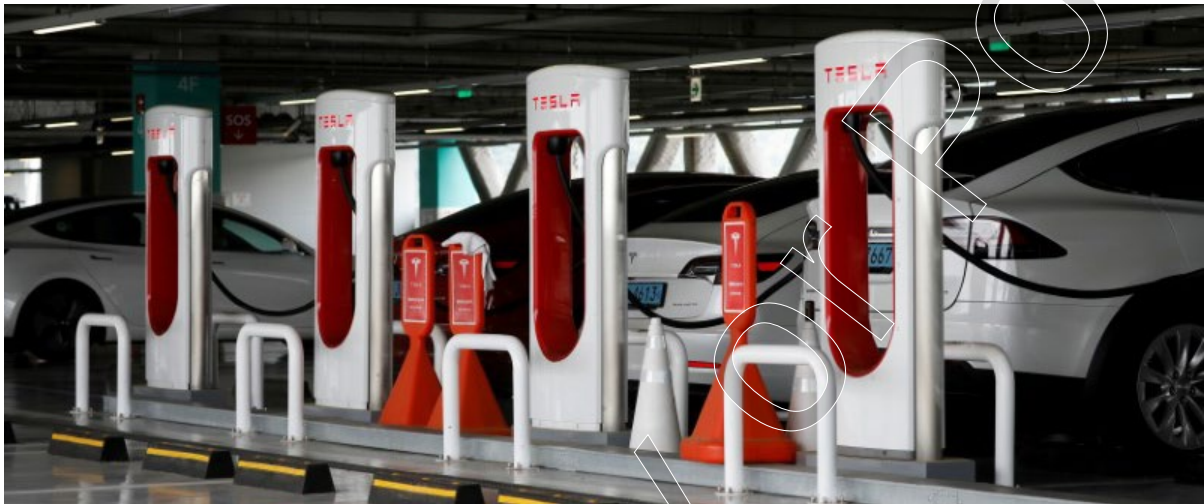
Hydrogen Fuel-Cell Electric Vehicle (FCEV) A hydrogen FCEV ran on an electric motor powered by a hydrogen gas fuel cell. FCEVs emitted water vapor and heat but no GHGs, fueled as quickly as ICE vehicles, and had a range of around 300 miles per tank.

Source: Electric Power Research Institute, cited in Auto Outlet USA, "What Are The Different Types Of Electric Vehicles," January 12, 2021, <https://bit.ly/3Wmbu2a>, accessed October 2022; U.S. Department of Energy, "Fuel Cell Electric Vehicles," Alternative Fuels Data Center, <https://bit.ly/3innr9p>, accessed January 2023.

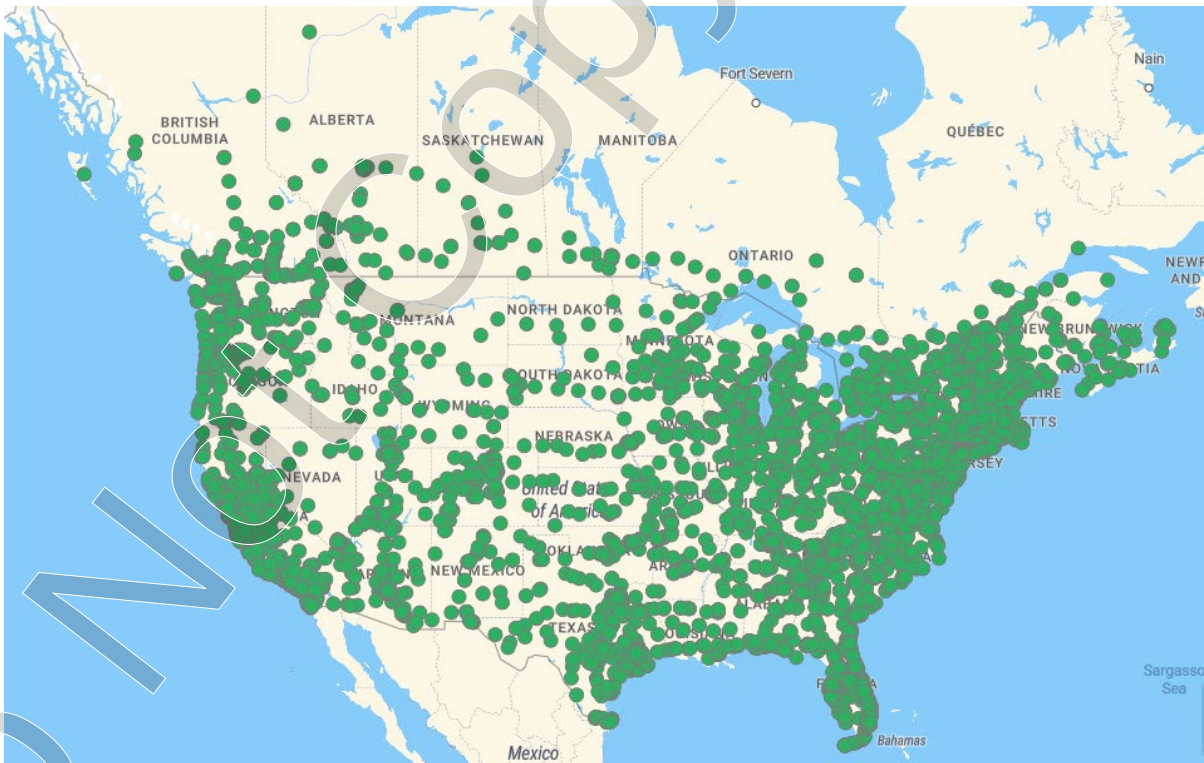
Exhibit 6 U.S. Car Production, by powertrain type, 1980 to 2020

	Gasoline	Diesel	HEV	EV	PHEV
1980	95.7%	4.3%	0.0%	0.0%	0.0%
1985	99.1%	0.9%	0.0%	0.0%	0.0%
1990	99.9%	0.1%	0.0%	0.0%	0.0%
1995	100.0%	0.0%	0.0%	0.0%	0.0%
2000	99.8%	0.1%	0.0%	0.0%	0.0%
2005	98.6%	0.3%	1.1%	0.0%	0.0%
2010	95.5%	0.7%	3.8%	0.0%	0.0%
2015	95.9%	0.9%	2.4%	0.5%	0.3%
2020	92.4%	0.5%	4.9%	1.8%	0.5%

Source: Environmental Protection Agency via Statista, Metals & Electronics, Vehicle Manufacturing Distribution of car production of the United States from 1980 to 2021, by powertrain, accessed November 2022.

Exhibit 7 Tesla Supercharger

In 2023, Tesla touted 40,000 Supercharger locations in North America, recharging for up to 200 miles at 250 kW in 25 minutes.



Source: Kim Hong-Ji in Reuters, "German minister in talks with Tesla over sharing supercharger network," June 16, 2021, <https://reut.rs/3GadHIC>; Tesla, "Supercharger," <https://bit.ly/3IDP0Uk>; and © MapTiler © OpenStreetMap contributors, "Electric Vehicle Charging Station Locations," U.S. Department of Energy, Energy Efficiency & Renewable Energy, Alternative Fuels Data Center, <https://bit.ly/3KQbc0a> and <http://openmaptiles.org/>, all accessed March 2023.

Exhibit 8 Tesla Stores

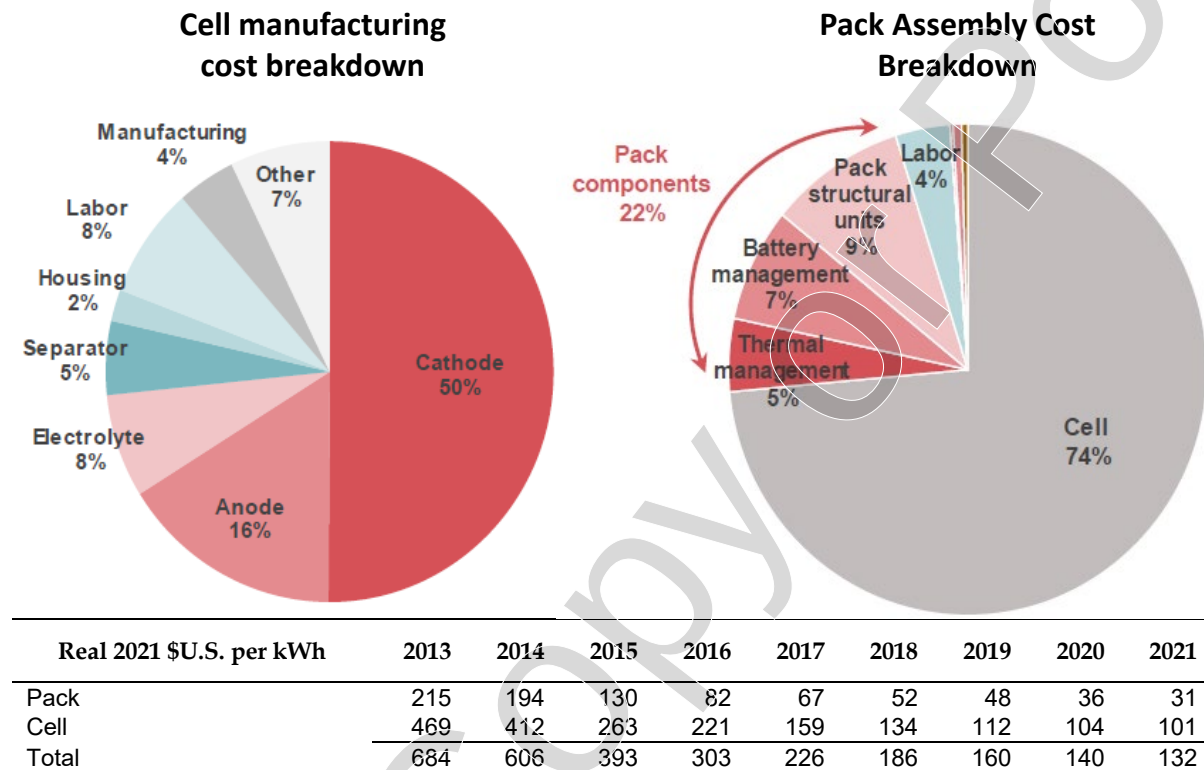
Source: Raysonho, "Tesla Motors at Yorkdale," TeslaMotorsYorkdale5.jpg, August 22, 2014, <https://bit.ly/43S6q9H>, available under the Creative Commons CC0 1.0 Universal Public Domain Dedication; Windell Oskay, "Tesla Visit 9," May 20, 2013, <https://bit.ly/40zGa0L>, available under the Creative Commons BY 2.0; both accessed April 2023.

Exhibit 9 ICE Vehicle Ban Timeline Across Different Countries

Government	Year	Vehicle Category	Target
Norway	2025	PCs, LCVs, urban uses	New vehicle sales 100% ZEV
Netherlands	2025	Urban buses	New vehicle purchases 100% ZEV
	2030	PCs	
Denmark	2030	PCs	No new gasoline or diesel vehicle sales
	2035		No new gasoline, diesel, or PHEV sales
Iceland	2030	PCs	No new gasoline or diesel vehicle registrations
Ireland	2030	PCs	No sales of new fossil fuel vehicles
Slovenia	2030	PCs, LCVs	No new registrations with CO ₂ emissions >50 grams/km
Sweden	2030	PCs	No sales of new gasoline or diesel vehicles
United Kingdom	2030	PCs, LCVs	No sales of new ICE vehicles. PHEVs allowed to 2035. Likely that non-PHEV allowed to 2035 after pressure from Japanese original equipment manufacturers
France	2040	PCs, LCVs	No sales of new fossil fuel vehicles
Spain	2040	PCs, LCVs	New vehicle sales 100% ZEV
Portugal	2040	All vehicles	No new ICE vehicle purchases
Germany	2050	PCs	New vehicle sales 100% ZEV
U.S.: California	2035	PCs, LCVs	New vehicle sales 100% ZEV
Colombia	2035	Public transport	New purchases 100% ZEV
Canada	2040	PCs, LCVs	New vehicle sales 100% ZEV
Costa Rica	2050	PCs, LCVs	New vehicle sales 100% ZEV
U.S.: Connecticut, Maryland, Massachusetts, New Jersey, New York, Oregon, Rhode Island, Vermont, Washington	2050	PCs	New vehicle sales 100% ZEV
U.S.: Connecticut, Colorado, Hawaii, Maine, Maryland, Massachusetts, New Jersey, New York, North Carolina, Oregon, Pennsylvania, Rhode Island, Vermont, Washington, Washington, DC	2050	Medium- and heavy-duty vehicles	New vehicle sales 100% ZEV
China: Hainan	2020	Gov't and car-sharing vehicles, light-duty trucks	New vehicle sales 100% electric
		Private cars	
	2030		
Japan	2035	PCs, LCVs	Ban of ICE vehicle sales
Singapore	2030	All vehicles	No ICE vehicles
Israel	2030	Private vehicles	New vehicle sales 100% electric
Cape Verde	2035	PCs, LCVs, buses, and trucks	No imports of ICE vehicles using fossil fuels

Source: Adapted by casewriter from Stephen Reitman and Steve Pereira Fernandes, "Automotive Sector: Coping with Disruption and Economic Uncertainties," Société Générale, September 2022, p. 43, via Refinitiv/AMR, accessed November 2022.

Note: PC = passenger cars, LCV = light commercial vehicle.

Exhibit 10 Falling Lithium-Ion Battery Cost

Source: Adapted by casewriter from Stephen Reitman and Steve Pereira Fernandes, "Automotive Sector: Coping with Disruption and Economic Uncertainties," Société Générale, September 2022, p. 92, via Refinitiv/AMR, accessed November 2022.

Note: Cell manufacturing and pack assembly costs breakdowns were modeled on a 60kWh battery pack assembled by 60Ah prismatic cells with NMC (622) cathode and natural graphite anode. The pack was assumed to be assembled in a Chinese plant of 10GWh annual manufacturing capacity and 50,000 packs with a 100% utilization rate and either a 5% scrappage rate (cell manufacturing) or 1% scrappage rate (pack assembly).

Exhibit 11a Tesla Financial Data, selected, FY2011 to FY2021

	FY2011	FY2012	FY2013	FY2014	FY2015	FY2016	FY2017	FY2018	FY2019	FY2020	FY2021
Revenue (\$U.S. billions)	0.2	0.4	2.0	3.2	4.1	7.0	11.8	21.5	24.6	31.5	53.8
1-Year Revenue Growth	74.9%	102.3%	387.2%	58.8%	26.5%	73.0%	68.0%	82.5%	14.5%	28.3%	70.7%
EBITDA/Revenue	-114.8%	-88.4%	2.2%	1.4%	-7.3%	4.3%	0.0%	7.6%	8.8%	13.5%	17.5%
EBIT/Revenue	-123.1%	-95.4%	-3.0%	-5.8%	-17.7%	-9.2%	-13.9%	-1.2%	0.3%	6.2%	12.1%
CAPEX/Revenue	90.2%	57.9%	13.1%	30.3%	40.4%	20.6%	34.7%	10.8%	5.8%	10.2%	12.1%
R&D/Revenue	102.3%	66.3%	11.5%	14.5%	17.7%	11.9%	11.7%	6.8%	5.5%	4.7%	4.8%
SG&A/Revenue	51.0%	36.4%	14.2%	18.9%	22.8%	20.1%	21.1%	13.2%	10.8%	10.1%	8.4%
Gross Profit/Revenue	30.2%	7.3%	22.7%	27.6%	22.8%	22.8%	18.9%	18.8%	16.6%	21.0%	25.3%
Gross Profit/Revenue by Line of Business											
Automotive Sales	53.7%	46.7%	99.5%	100.1%	85.7%	82.6%	81.5%	97.6%	84.0%	74.0%	86.1%
Development Services	46.3%	53.3%	0.5%	-0.1%	-	-	-	-	-	-	-
Automotive Leasing	-	-	-	-	13.6%	17.5%	17.9%	9.8%	10.1%	7.4%	12.0%
Energy Generation & Storage	-	-	-	-	0.2%	0.2%	10.9%	4.7%	4.7%	0.3%	-0.9%
Services & Other	-	-	-	-	0.4%	-0.3%	-10.3%	-12.1%	-13.4%	-5.5%	-0.8%
Automotive Regulatory Credits	-	-	-	-	-	-	-	-	14.6%	23.8%	10.8%
Total Vehicles Produced	-	3,100	-	-	30,240	84,000	100,800	254,530	367,656	509,737	930,422
Total Vehicles Delivered	-	2,650	22,477	56,768	50,580	76,230	103,184	245,506	365,232	499,647	936,222

Source: Capital IQ, Inc., a Standard and Poor's company, accessed November 2022 and January 2023; Tesla, Inc. 10-K filings, accessed January 2023.

Note: Dash = not reported or not available.

Exhibit 11b Tesla and Selected Competitors' Financial Ratios, FY2021

	Tesla	General Motors	Volkswagen	Ford	Honda	Toyota	Mercedes
Gross Profit/Revenue	25.3%	14.2%	17.8%	12.1%	20.7%	17.8%	21.8%
EBITDA/Revenue	17.5%	13.3%	12.6%	11.9%	14.0%	13.0%	14.4%
EBIT/Revenue	12.1%	9.0%	8.0%	7.8%	5.0%	8.1%	11.2%
CAPEX/Revenue	12.1%	17.4%	4.3%	4.6%	2.4%	12.8%	3.4%
R&D/Revenue	4.8%	-	-	-	5.6%	-	4.0%
SG&A/Revenue	8.4%	5.2%	10.4%	4.3%	10.1%	9.7%	7.6%
Growth Over Prior Year	70.7%	3.7%	12.3%	7.2%	-11.8%	-8.9%	9.9%
Price/Earnings	216	8.7	8.7	4.7	8.7	10.7	3.1
Price/Book Value	36	1.5	1.0	1.8	0.6	1.0	1.0

Source: Capital IQ, Inc., a Standard and Poor's company, accessed November 2022.

Note: Dash = not reported or not available.

Exhibit 11c Tesla % of Overall Sales, by Lines of Business and Geographic Segment, FY2011 to FY2021

	FY2011	FY2012	FY2013	FY2014	FY2015	FY2016	FY2017	FY2018	FY2019	FY2020	FY2021
% revenue from Automotive sales	72.7%	93.3%	95.4%	89.9%	84.8%	79.8%	72.6%	82.2%	78.8%	78.0%	82.0%
% revenue from Regulatory credits	-	-	-	-	-	-	-	-	2.4%	5.0%	2.7%
% revenue from Energy gen & storage	-	-	-	0.1%	0.4%	2.6%	9.5%	7.2%	6.2%	6.3%	5.2%
% revenue from Development services	27.3%	6.7%	-	-	-	-	-	-	-	-	-
% revenue from Services and other	-	-	4.6%	5.9%	7.2%	6.7%	8.5%	6.5%	9.1%	7.3%	7.1%
	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
% revenue from United States	-	-	-	-	-	60.0%	52.9%	69.3%	51.5%	48.2%	44.5%
% revenue from China	-	-	-	14.9%	-	15.2%	17.2%	8.2%	12.1%	21.1%	25.7%
% revenue from Other	-	15.5%	15.8%	26.2%	-	15.6%	20.0%	22.5%	36.4%	30.7%	29.7%
% revenue from Netherlands	-	-	-	-	-	4.4%	2.8%	-	-	-	-
% revenue from Norway	-	1.0%	10.8%	12.9%	-	4.8%	7.0%	-	-	-	-
% revenue from Americas/North America	53.5%	83.5%	73.5%	46.0%	-	-	-	-	-	-	-
% revenue from Europe	41.3%	-	-	-	-	-	-	-	-	-	-
% revenue from Asia	5.2%	-	-	-	-	-	-	-	-	-	-
	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Source: Capital IQ, Inc., a Standard and Poor's company, accessed November 2022 and January 2023.

Note: Dash = not available or not reported; changing geographies reflect changing reporting structure, totaling to 100% in each year.

Exhibit 12a Range, Efficiency, Prices for Selected EV Models, 2022

Brand	Model	Battery (kWh)	% of battery used*	Range (km)	Consumption* (kWh/100km)	Price (€)	Fastcharge max power (kW)
Tesla	Model 3 (LR2)	77.0	91%	580	12.1 - 14.8	53,560	190
Tesla	Model S (LR)	95.0	95%	663	13.6	86,990	250
Tesla	Model X (LR)	95.0	95%	580	15.5	95,990	250
Tesla	Model Y (LR2)*	75.0	97%	505	14.4	58,620	250
Audi	e55 Quattro	95.0	91%	436	19.8 - 22.4	79,445	155
Audi	eeGT Quattro	93.4	91%	487	17.5 - 19.9	99,800	270
Audi	Q4 e-tron 50	82.0	94%	488	17.9 - 15.8	52,900	126
Porsche	Taycan Turbo	93.4	90%	452	18.5 - 22.9	153,016	262
Porsche	Taycan 4S	79.2	90%	408	17.4 - 21.1	106,487	225
Volkswagen	ID.3 Pro S	82.0	94%	549	13.5 - 14.0	41,995	126
Volkswagen	ID.4 Pro.*	82.0	94%	522	14.8 - 17.1	48,691	126
BMW	i3 (120 Ah)	42.2	90%	308	12.3 - 15.3	39,000	49
BMW	i4*	80.0	93%	590	13.3	69,540	150
BMW	iX3	80.0	93%	460	16.1 - 18.5	66,300	155
BMW	iX xDrive 50	110.0	95%	629	17.5 - 20.0	98,000	200
Mini	Cooper SE	32.6	89%	234	12.4 - 15.2	32,500	49
Mercedes	EQC	85.0	94%	411	19.5 - 22.3	66,069	112
Mercedes	EQB*	70.0	95%	419	15.9 - 18.1	60,000	100
Mercedes	EQS 450+*	112.0	96%	780	14.0 - 15.7	106,374	200
Jaguar	I-Pace EV400	90.0	94%	470	18.0 - 22.0	77,300	104
Renault	Zoe (R110)	54.7	95%	395	13.2 - 17.4	31,990	46
Peugeot	e-208	50.0	90%	339	13.3 - 16.4	30,450	99
Peugeot	e-2008	50.0	90%	331	13.6	35,450	99
Opel	Corsa-e	50.0	90%	330	13.6 - 17.0	29,000	99
Nissan	Leaf e+	62.0	90%	385	14.5 - 18	38,350	100
Kia	Niro	67.5	95%	455	14.1 - 15.9	39,090	77
Hyundai	Kona	67.5	95%	484	13.2 - 14.7	41,850	77
Hyundai	Ioniq 5 (LR)*	72.6	95%	395	18.4	55,000	232

Source: Adapted from Stephen Reitman and Steve Pereira Fernandes, "Automotive Sector: Coping with Disruption and Economic Uncertainties," Société Générale, September 2022, p. 82, via Refinitiv/AMR, accessed November 2022.

Note: * Based on Worldwide Harmonised Light Vehicles Test Procedure (WLTP) data, or estimates where WLTP was not available. Larger battery packs increased range. However, there were diminishing returns, as larger batteries added weight and therefore reduced overall efficiency (kWh/100km). Another factor which influenced range was the percentage of the battery that was actually used. Using a high percentage of the battery's capacity reduced the number of times it could be cycled (charged/discharged) and potentially increased the risk of spontaneous fires. Manufacturers left a buffer or margin of safety between the stated battery capacity and what was actually used. Legacy manufacturers tended to err on the side of caution and leave a larger buffer; this partly explained the lower ranges they achieved versus competing Tesla products. However, as the legacy car companies' know-how and confidence in the technology increased, the efficiency and range gaps were narrowing. LR = long range; LR2 = long range dual motor.

Exhibit 12b Comparative Lifecycle GHG Emissions, tCO₂e per vehicle lifetime, by emissions source

	Vehicle manufacturing	Batteries: assembly & other	Batteries: minerals	Electricity	Fuel cycle (well-to-wheel)
BEV Base Case	5.4	1.2	1.4	11.7	0
BEV High-GHG Minerals Case	5.4	1.2	2.8	11.7	0
ICE Vehicle	6	0.01	0.03	0	35.9

Source: IEA, Comparative life-cycle greenhouse gas emissions of a mid-size BEV and ICE vehicle, updated October 26, 2022, license: CC BY 4.0, <https://bit.ly/3GfVKbt>, accessed November 2022.

Note: tCO₂e was tons of CO₂ equivalent, a standardized measure of GHG emissions. The “High-GHG minerals” case assumed double emissions intensity for battery minerals (70 kgCO₂e/kWh compared to 35 kgCO₂e/kWh in the base case; other assumptions were the same). The values were for a vehicle manufactured from manufacturing lines assuming dynamic global average grid carbon intensity (including transmissions, distribution and charging losses, weighted for mileage decay over a 20-year lifetime). The ranges shown for BEV represent cases for charging with a static low-carbon (50 gCO₂e/kWh) and high-carbon electricity mix (800 gCO₂e/kWh). Vehicle assumptions: 200,000 km lifetime mileage; ICE fuel economy 6.8 liters of gasoline-equivalent per 100 km; BEV fuel economy 0.19 kWh/km; BEV battery 40 kWh nickel manganese cobalt in a 6:2:2 ratio.

Exhibit 13 Auto Manufacturing MSCI's ESG Analysis Framework

Environmental		Social		Governance	
Issue	Average Weight	Issue	Average Weight	Issue	Average Weight
Product Carbon Footprint	15.8%	Product Safety & Quality	20.3%	Governance	33.8%
Opportunities in Clean Tech	13.0%	Labor Management	16.7%	Ownership & Control	-
Toxic Emissions & Waste	0.2%	Chemical Safety	0.3%	Board	-
Packaging Materials & Waste	-	Consumer Finance Protection	-	-	-
Opportunities in Green Building	-	Controversial Sourcing	-	Accounting	-
Raw Material Sourcing	-	Opportunities in Nutrition & Health	-	Business Ethics	-
Climate Change Vulnerability	-	Insuring Health & Demographic Risk	-	Tax Transparency	-
Financing Environmental Impact	-	Responsible Investment	-		
Biodiversity & Land Use	-	Community Relations	-		
Opportunities in Renewable Energy	-	Access to Communications	-		
Carbon Emissions	-	Access to Finance	-		
Water Stress	-	Access to Health Care	-		
Electronic Waste	-	Privacy & Data Security	-		
		Supply Chain Labor Standards	-		
		Health & Safety	-		
		Human Capital Development	-		

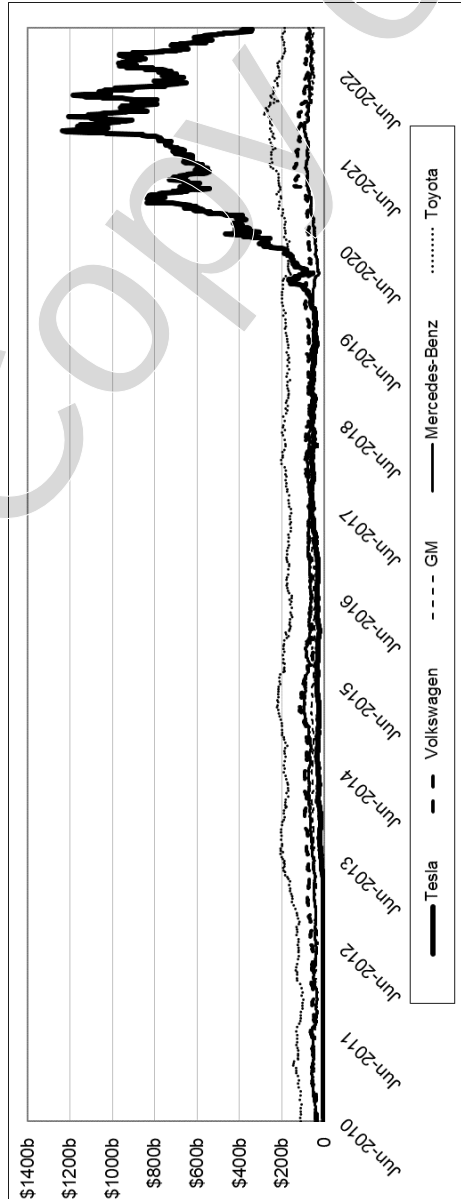
Source: Adapted by casewriter from MSCI, ESG Industry Materiality Map, Consumer Discretionary, Automobile Manufacturers, <https://bit.ly/3HrDBYL>, accessed December 2022.

Note: Dash indicated unavailable.

Exhibit 14a Tesla and Selected Competitors' Market Capitalization, \$U.S. billions, 2022 Peak and December 2022

	2022 Peak Market Cap	December 2022 Market Cap
Tesla	1,240	389
Toyota	292	187
Volkswagen	144	68
Mercedes-Benz Group	103	66
GM	96	48

Source: Capital IQ, Inc., a division of Standard and Poor's, accessed January 2023.

Exhibit 14b Tesla and Selected Competitors' Market Capitalization, \$U.S. billions, June 2010 to January 2023

Source: Capital IQ, Inc., a division of Standard and Poor's, accessed January 2023.

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²³⁹ “Frustrated Tesla investors call on Elon Musk to step down as CEO,” *Drive*, December 21, 2022, <https://bit.ly/3Z210Gy>, accessed January 2023.

²⁴⁰ “Tesla’s robot waves but can’t walk, yet. Musk plans to make millions of them.” Reuters, October 2, 2022, <https://shorturl.at/hrtGV>, accessed January 2023.

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